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Display panel and display device

Abstract

A display panel and a display device are provided. The display panel includes: a substrate; light-emitting elements on the substrate; and a light-blocking structure on one light-emitting element. One light-emitting element includes a first light-emitting element and a second light-emitting element. An area of an orthographic projection of the first light-emitting element is S1, and an area of an orthographic projection of one second light-emitting element is S2. One light-blocking structure includes a first light-blocking structure and a second light-blocking structure. An orthographic projection of one first light-blocking structure at least partially surrounds one corresponding first light-emitting element and a distance from the first light-blocking structure to the first light-emitting element is d1. An orthographic projection of one second light-blocking structure at least partially surrounds one corresponding second light-emitting element and a distance from the second light-blocking structure to the second light-emitting element is d2. $|S1-S2|>0$, and $|d1-d2|>0$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority of Chinese Patent Application No. 202210906565.0, filed on Jul. 29, 2022, the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure generally relates to the field of display technologies and, more particularly, relates to a display panel and a display device.

BACKGROUND

(3) With the continuous development of display technology, the manufacturing technology of display panels is also becoming mature. Existing display panels mainly include liquid crystal display (LCD), organic light-emitting Diode (OLED), and so on. An organic light-emitting display device, which is a self-luminous display device, does not require a separate light source, can

operate at a low voltage, is lightweight and thin, and provides high-quality characteristics including self-luminescence, a wide viewing angle, high contrast, and fast response. Therefore, organic light-emitting display devices have been attracting attention as next-generation display devices. A general organic light-emitting display device is a self-emission type display device including organic light-emitting elements (OLEDs), and one OLED includes a hole injection electrode, an electron injection electrode, and an organic emission layer between the hole injection electrode and the electron injection electrode. In the organic light-emitting display device, holes injected from the hole injection electrode and electrons injected from the electron injection electrode may combine in the organic emission layer to generate excitons, and the excitons may transit from an excited state to a ground state and generate light, realizing a picture show.

(4) With the development of display technology, the lifespan of luminescent materials, the functions undertaken by the regions, and the luminous color of different sub-pixels on the display panel have different effects on the display effect. Therefore, the white light characteristics of the display panel under different viewing angles shows obvious difference. It is easy to affect the uniformity of the white light characteristics of the display panel under different viewing angles and affect the display quality.

(5) Therefore, to provide a display panel and a display device capable of improving the difference in white light characteristics under different viewing angles, therefore improving the uniformity of white light characteristics under different viewing angles and display quality, is a technical problem to be solved.

SUMMARY

(6) One aspect of the present disclosure provides a display panel. The display panel includes: a substrate; light-emitting elements on a side of the substrate; and a light-blocking structure on a side of one light-emitting element away from the substrate. One light-emitting element includes a first light-emitting element and a second light-emitting element. An area of an orthographic projection of one first light-emitting element on the substrate is S1, and an area of an orthographic projection of one second light-emitting element on the substrate is S2. One light-blocking structure includes a first light-blocking structure and a second light-blocking structure. An orthographic projection of one first light-blocking structure on the substrate at least partially surrounds one corresponding first light-emitting element and a distance from the first light-blocking structure to the corresponding first light-emitting element along a first direction is d1. An orthographic projection of one second light-blocking structure on the substrate at least partially surrounds one corresponding second light-emitting element and a distance from the second light-blocking structure to the corresponding second light-emitting element along the first direction is d2. The first direction is parallel to a plane where the display panel is located; and $|S1-S2|>0$, and $|d1-d2|>0$.

(7) Another aspect of the present disclosure provides a display device. The display device includes a display panel. The display panel includes a substrate; light-emitting elements on a side of the substrate; and a light-blocking structure on a side of one light-emitting element away from the substrate. One light-emitting element includes a first light-emitting element and a second light-emitting element. An area of an orthographic projection of one first light-emitting element on the substrate is S1, and an area of an orthographic projection of one second light-emitting element on the substrate is S2. One light-blocking structure includes a first light-blocking structure and a second light-blocking structure. An orthographic projection of one first light-blocking structure on the substrate at least partially surrounds one corresponding first light-emitting element and a distance from the first light-blocking structure to the corresponding first light-emitting element along a first direction is d1. An orthographic projection of one second light-blocking structure on the substrate at least partially surrounds one corresponding second light-emitting element and a distance from the second light-blocking structure to the corresponding second light-emitting element along the first direction is d2. The first direction is parallel to a plane where the display panel is located; and $|S1-S2|>0$, and $|d1-d2|>0$.

(8) Other aspects or embodiments of the present disclosure can be understood by those skilled in the art in light of the description, the claims, and the drawings of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following drawings are merely examples for illustrative purposes according to various disclosed embodiments and are not intended to limit the scope of the present disclosure.
- (2) FIG. 1 illustrates a planar structure of an exemplary display panel consistent with various disclosed embodiments in the present disclosure;
- (3) FIG. 2 illustrates a locally enlarged view of a M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure;
- (4) FIG. 3 illustrates a cross-sectional view of a region of the display panel in FIG. 2 along an AA' direction, consistent with various disclosed embodiments in the present disclosure;
- (5) FIG. 4 illustrates an enlarged view of the M1 region of the display panel in FIG. 1, consistent with various disclosed embodiments in the present disclosure;
- (6) FIG. 5 illustrates another exemplary planar structure of a display panel consistent with various disclosed embodiments in the present disclosure;
- (7) FIG. 6 illustrates a locally enlarged view of a M2 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (8) FIG. 7 illustrates a locally enlarged view of a M3 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (9) FIG. 8 illustrates a locally enlarged view of a M4 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (10) FIG. 9 illustrates another locally enlarged view of the M4 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (11) FIG. 10 illustrates another locally enlarged view of the M4 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (12) FIG. 11 illustrates a locally enlarged view of an M5 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (13) FIG. 12 illustrates another locally enlarged view of the M5 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (14) FIG. 13 illustrates another locally enlarged view of the M4 region in the display panel in FIG. 5 consistent with various disclosed embodiments in the present disclosure;
- (15) FIG. 14 illustrates a cross-sectional view of the M4 region in the display panel in FIG. 5 along a B-B' direction, consistent with various disclosed embodiments in the present disclosure;
- (16) FIG. 15 illustrates another locally enlarged view of the M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure;
- (17) FIG. 16 illustrates a cross-sectional view of the display panel in FIG. 15 along a C-C' direction, consistent with various disclosed embodiments in the present disclosure;
- (18) FIG. 17 illustrates another locally enlarged view of the M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure;
- (19) FIG. 18 illustrates a cross-sectional view of the display panel in FIG. 17 along a D-D' direction, consistent with various disclosed embodiments in the present disclosure;
- (20) FIG. 19 illustrates another locally enlarged view of the M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure;
- (21) FIG. 20 illustrates a cross-sectional view of the display panel in FIG. 19 along an E-E' direction, consistent with various disclosed embodiments in the present disclosure;
- (22) FIG. 21 illustrates another cross-sectional view of the display panel in FIG. 19 along an E-E'

direction, consistent with various disclosed embodiments in the present disclosure;

(23) FIG. 22 illustrates another cross-sectional view of the display panel in FIG. 19 along an E-E' direction, consistent with various disclosed embodiments in the present disclosure;

(24) FIG. 23 illustrates another locally enlarged view of the M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure;

(25) FIG. 24 illustrates a cross-sectional view of the display panel in FIG. 19 along an F-F' direction, consistent with various disclosed embodiments in the present disclosure;

(26) FIG. 25 illustrates another locally enlarged view of the M1 region in the display panel in FIG. 1 consistent with various disclosed embodiments in the present disclosure; and

(27) FIG. 26 illustrates an exemplary display device consistent with various disclosed embodiments in the present disclosure.

DETAILED DESCRIPTION

(28) Reference will now be made in detail to exemplary embodiments of the disclosure, which are illustrated in the accompanying drawings. Hereinafter, embodiments consistent with the disclosure will be described with reference to drawings. In the drawings, the shape and size may be exaggerated, distorted, or simplified for clarity. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts, and a detailed description thereof may be omitted.

(29) Further, in the present disclosure, the disclosed embodiments and the features of the disclosed embodiments may be combined under conditions without conflicts. It is apparent that the described embodiments are some but not all of the embodiments of the present disclosure. Based on the disclosed embodiments, persons of ordinary skill in the art may derive other embodiments consistent with the present disclosure, all of which are within the scope of the present disclosure.

(30) Moreover, the present disclosure is described with reference to schematic diagrams. For the convenience of descriptions of the embodiments, the cross-sectional views illustrating the device structures may not follow the common proportion and may be partially exaggerated. Besides, those schematic diagrams are merely examples, and not intended to limit the scope of the disclosure. Furthermore, a three-dimensional (3D) size including length, width, and depth should be considered during practical fabrication.

(31) FIG. 1 illustrates a top view of an exemplary display panel according to one embodiment of the present disclosure, FIG. 2 illustrates a locally enlarged view of a M1 region in the display panel shown in FIG. 1, FIG. 3 illustrates a cross-sectional view of the display panel in FIG. 2 along an AA' direction. As shown in FIG. 1 to FIG. 3, in one embodiment, the display panel 100 may include a substrate 10, light-emitting elements 20 on a side of the substrate 10, light-blocking structures 30 at a side of the light-emitting elements 20 away from the substrate 10.

(32) One light-emitting element 20 may include a first light-emitting element 201 and a second light-emitting element 202. An area of an orthographic projection of the first light-emitting element 201 on the substrate 10 may be S1, and an area of an orthographic projection of the second light-emitting element 201 on the substrate 10 may be S2.

(33) One light-blocking structure 30 may include a first light-blocking structure 301 and a second light-blocking structure 302. An orthographic projection of the first light-blocking structure 301 of one light-blocking structure 30 on the substrate 10 may at least partially surround the first light-emitting element 201 of one corresponding light-emitting element 20. Along a first direction X, a distance from the first light-blocking structure 301 to the corresponding first light-emitting element 201 may be d1.

(34) An orthographic projection of one second light-blocking structure 302 on the substrate 10 may at least partially surround one corresponding second light-emitting element 202, and along the first direction X, a distance from the second light-blocking structure 302 to the corresponding second light-emitting element 202 may be d2. The first direction X may be parallel to the plane where the display panel 000 is located.

$|S1-S2|>0$, and $|d1-d2|>0$.

(35) The display panel **000** may be an organic light-emitting diode display panel. The display panel **000** may include the substrate **10** on which other film structures of the display panel **000** are fabricated. The light-emitting elements **20** may be disposed on one side of the substrate **10**. It can be understood that one light-emitting element **20** of the present disclosure may be understood as at least a portion of an area corresponding to one sub-pixel of the display panel **000**. Optionally, a driving circuit layer **40** may be disposed on one side of the substrate **10**, and the driving circuit layer **40** may be used to fabricate a circuit structure for driving the light-emitting elements **20** to emit light, such as pixel circuits including transistors, etc. Driving signals may be provided to each light-emitting element **20** corresponding to each sub-pixel through the circuit structure in the driving circuit layer **40**. An anode layer **50**, a pixel definition layer **60** and a cathode layer **70** may be provided on a side of the driving circuit layer **40** away from the substrate **10**. The pixel definition layer **60** may be provided with openings **60K** penetrating through the thickness of the pixel definition layer **60**, to form opening areas **600** of the pixel definition layer **60**. One opening **60K** of the pixel definition layer **60** may be used to accommodate an organic light-emitting material of one corresponding light-emitting element **20**. The organic light-emitting material may be driven by the electric field formed by the anode layer **50** and the cathode layer **70**, to emit light through injection and combination of carriers. Therefore, the light-emitting element **20** of one corresponding sub-pixel may emit light.

(36) One light-emitting element **20** may include one first light-emitting element **201** and one second light-emitting element **202**. The area of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be $S1$, and the area of the orthographic projection of the second light-emitting element **201** on the substrate **10** may be $S2$. The area $S1$ of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be understood as an area of an orthographic projection of one first opening area **600A** in the pixel definition layer **60** corresponding to the first light-emitting element **201**, and the area $S2$ of the orthographic projection of the second light-emitting element **202** on the substrate **10** may be understood as an area of an orthographic projection of one second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202**.

(37) The display panel **000** may further include the light-blocking structures **30** at the side of the light-emitting elements **20** away from the substrate **10**. The light-blocking structures **30** may be disposed between different light-emitting elements **20** for blocking, to avoid light crosstalk between different light-emitting elements **20**. Optionally, the light-blocking structures **30** may be located on the side of the cathode layer **70** away from the substrate **10**. In some other optional embodiments, the light-blocking structures **30** may also be located in other film layers on the side of the light-emitting elements **20** away from the substrate **10**. The present disclosure has no limit on this.

(38) One light-blocking structure **30** may include one first light-blocking structure **301** and one second light-blocking structure **302**. In one light-blocking structure **30**, the first light-blocking structure **301** may correspond to one first light-emitting element **201**. The orthographic projection of the first light-blocking structure **301** may at least partially surround the corresponding first light-emitting element **201**. In one embodiment, the orthographic projection of the first light-blocking structure **301** may surround the corresponding first light-emitting element **201**, as shown in FIG. 2. In one light-blocking structure **30**, the second light-blocking structure **302** may correspond to one second light-emitting element **202**. The orthographic projection of the second light-blocking structure **302** may at least partially surround the corresponding second light-emitting element **202**. In one embodiment, the orthographic projection of the second light-blocking structure **302** may surround the corresponding second light-emitting element **202**, as shown in FIG. 2. That is, the first light-blocking structure **301** may be at least provided at a portion of the periphery of the first light-emitting element **201**, and the second light-blocking structure **302** may be at least provided at a

portion of the periphery of the second light-emitting element **202**. It can be understood that the present disclosure does not specifically limit the material of the light-blocking structures **30**, and the material only needs to be opaque and be able to play a light-blocking role.

(39) With the development of display technology, the lifespan of the light-emitting materials of different light-emitting elements on the display panel, the functions undertaken by the regions where different light-emitting elements are located, and the light-emitting colors of different light-emitting elements have different effects on the display effect. The white light characteristics of the panels at specific viewing angles are different, which in turn easily affects the uniformity of the white light characteristics of the display panel at different viewing angles and affects the display quality of the display panel. To solve the above problems, the display panel provided by the present disclosure may adjust different light-emitting elements to different degrees according to the display differences of different light-emitting elements, to ensure that the final display effect of the display panel is uniform and good. In existing technologies, the area of a general light-emitting element, that is, the area of the corresponding opening area of the pixel definition layer in the present disclosure, and the design of the light-blocking structures, are two factors that may significantly affect the values of relative brightness of each sub-pixel after being shielded by one corresponding light-blocking structure, the change of brightness and color coordinate of each sub-pixel under a specific viewing angle.

(40) In the display panel **000** provided by the present disclosure, along the first direction X, a distance from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be $d1$. A distance from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be $d2$. $|S1-S2|>0$, and $|d1-d2|>0$. That is, the area $S1$ of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area $S2$ of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance $d1$ between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance $d2$ between the second light-blocking structure **302** and the second light-emitting element **202**.

(41) Optionally, in one embodiment shown in FIG. 1 and FIG. 2, the area $S1$ of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be larger than the area $S2$ of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be larger than the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance $d1$ between the first light-blocking structure **301** and the first light-emitting element **201** may be smaller than the distance $d2$ between the second light-blocking structure **302** and the second light-emitting element **202**. In another embodiment shown in FIG. 1 and FIG. 4 which is another enlarger view of the M1 region in FIG. 1, the area $S1$ of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be larger than the area $S2$ of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be larger than the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance $d1$ between the first light-blocking structure **301** and the first light-emitting element **201** may be larger than the distance $d2$ between the second light-blocking

structure **302** and the second light-emitting element **202**. The present disclosure has no limit on this, and it may be configured according to actual requirements, as long as the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**, and the distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d2 between the second light-blocking structure **302** and the second light-emitting element **202** along the first direction X.

(42) The first direction X may be any direction parallel to the plane where the display panel **000** is located, and the present disclosure has no limit on this. The distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** along the first direction X may be a minimum distance between the first light-blocking structure **301** and the first light-emitting element **201** along the first direction, that is, a minimum of distances between a side edge of the first light-blocking structure **301** close to the corresponding first light-emitting element **201** (which may also be a point, as shown in FIG. 2, according to different shape of the first light-blocking structure) and a side edge of the first light-emitting element **201** close to the corresponding first light-blocking structure **301** (which may also be a point, as shown in FIG. 2, according to different shape of the first light-emitting element). The distance d2 between the second light-blocking structure **302** and the second light-emitting element **202** along the second direction X may be a minimum distance between the second light-blocking structure **302** and the second light-emitting element **202** along the second direction, that is, a minimum of distances between a side edge of the second light-blocking structure **302** close to the corresponding second light-emitting element **202** (which may also be a point, as shown in FIG. 2, according to different shape of the second light-blocking structure) and a side edge of the second light-emitting element **202** close to the corresponding second light-blocking structure **302** (which may also be a point, as shown in FIG. 2, according to different shape of the second light-emitting element).

(43) For description purposes only, the above embodiments where an orthographic projection of one light-emitting element **20** on the substrate **10** is a circle is used as examples to illustrate the present disclosure, and does not limit the scope of the present disclosure. In various embodiments, an orthographic projection of one light-emitting element **20** on the substrate **10** may be another suitable shape, and may be configured according to actual requirements. In one embodiment, an orthographic projection of one light-emitting element **20** on the substrate **10** may be a circle, and one light-blocking structure **30** corresponding to the light-emitting element **20** may at least partially surround the light-emitting element **20**. Specifically, in one embodiment, the orthographic projection of the light-blocking structure **30** on the substrate **10** may be a ring surrounding the light-emitting element **20**. In one embodiment, the orthographic projection of the light-blocking structure **30** on the substrate **10** may be an annular shape, to form the light-blocking structure **30** surrounding the light-emitting element **20**. In specific implementation, the shape of the orthographic projection of the light-blocking structure **30** on the substrate **10** may include but is not limited to this embodiment, and other shapes may also be included, which are not limited in the present disclosure. During specific implementation, the shape of the orthographic projection of the light-blocking structure **30** on the substrate **10** may be set according to actual requirements.

(44) In the present disclosure, the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d2 between the second light-blocking structure **302** and the second

light-emitting element **20**. Therefore, for the light-emitting elements **20** with different areas, for example, for a light-emitting element **20** with a larger area, the distance between one light-emitting element **20** and its corresponding light-blocking structure **30** in the direction parallel to the plane of the display panel **000** may be designed to be larger or smaller. Correspondingly, although the areas of the light-emitting elements **20** in the display panel **000** are different, by adjusting the distance between one light-emitting element **20** and its corresponding light-blocking structure **30** in the direction parallel to the plane of the display panel **000**, the relative brightness of two light-emitting elements **20** with different areas after being blocked by corresponding light-blocking structures **30** respectively may be kept consistent as much as possible, and the light-emitting brightness at a specific viewing angle may be also as consistent as possible. The display difference between different light-emitting elements **20** may be reduced, to improve the view angle difference of the white light of the display panel **000**. The uniformity of white light characteristics of the display panel **000** under different viewing angles may be improved, thereby helping to improve the display quality of the display panel **000**.

(45) The embodiments shown in the drawings are used as examples only to show structures that could be included in the display panel **000**. The structure of the display panel **000** may not be limited to these, and the display panel **000** may include other structures that are able to realize display function. The reference may be made to the structure of the organic light-emitting diode display panel in existing technologies, and will not be repeated here.

(46) In some other embodiment, the display panel **000** may be a micro LED display panel or a mini LED display panel. Correspondingly, one light-emitting element **20** may be any one of a micro LED or a mini LED (not shown in the figures). Because of different service lives of micro LEDs or mini LEDs with different colors, generally when a micro LED display panel or a mini LED display panel is fabricated, the orthographic projections of the micro LEDs or the mini LEDs of different colors on the substrate **10** may have different areas. At this time, to solve the problem of different white light characteristics at a specific viewing angle caused by the different effects of the luminous color of different light-emitting elements on the display effect, based on the display difference of different micro LEDs or mini LEDs, the different micro LED or mini LED may be adjusted to different degrees, to ensure that the final display effect of the display panel is uniform and good. Optionally, in the micro LED or mini LED display panel, generally the blue micro LEDs or mini LEDs have longer lives, and an orthographic projection area of one red micro LED on the substrate **10** (it can be understood that the area S1 of the first light-emitting element **101** in FIG. 2) may be set to be larger than the area of the orthographic projection of one blue micro LED on the substrate **10** (it can be understood as the area S2 of the second light-emitting element **102** in FIG. 2), and the area of the orthographic projection of one green micro LED on the substrate **10** may be also set to be larger than the area of the orthographic projection of one blue micro LED on the substrate **10**. At this time, along the first direction X, one distance between one red micro LED and its corresponding light-blocking structure may be configured to be different from the distance between one blue micro LED and its corresponding light-blocking structure. Further optional, along the first direction X, one distance between one red micro LED and its corresponding light-blocking structure may be configured to be smaller than the distance between one blue micro LED and its corresponding light-blocking structure. Similarly, the area of the orthographic projection of the green micro LED on the substrate **10** (which can be understood as the area S1 of the first light-emitting element **101** in FIG. 2) may be also set to be larger than the area of the orthographic projection of the blue micro LED on the substrate **10** (which can be understood as the area S2 of the second light-emitting element **102** in FIG. 2). At this time, along the first direction X, the distance between the green micro LED and its corresponding light-blocking structure may be set to be different from the distance between the blue micro LED and its corresponding light-blocking structure. Further optionally, in one embodiment, along the first direction X, the distance between the green micro LED and its corresponding light-blocking structure may be set to be smaller than

the distance between the blue micro LED and its corresponding light-blocking structure. It can be understood that, when the display panel is a micro LED or a mini LED display panel, the distance between one light-emitting element and its corresponding light-blocking structure in the first direction may be understood as a distance between one edge of an orthographic projection of one micro-LED or one mini LED on the substrate **10** and a nearest edge of its corresponding light-blocking structure. It should be noted that the present disclosure does not describe the structure of the differential design when the display panel is a micro-LED or sub-millimeter LED display panel, and can be understood by referring to the descriptions in the above embodiments.

(47) In some embodiments, as shown in FIG. 1 to FIG. 3, the area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10**. That is, the area of the orthographic projection of the first opening area **600A** in the pixel definition layer **60** corresponding to the first light-emitting element **201** on the substrate **10** may be larger than the area of the orthographic projection of the second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction, the distance d1 between the first light-blocking structure **301** and the corresponding first light-emitting element **201** may be smaller than the distance d2 between the second light-blocking structure **302** and the corresponding second light-emitting element **202**. That is, $S1 > S2$ and $d1 < d2$.

(48) In the present embodiment, when the area of the orthographic projection of one light-emitting element **20** in the display panel **000** on the substrate **10** is larger, the opening area **600** of the pixel definition layer **60** corresponding to the light-emitting element **20** may be also larger.

Correspondingly, along the first direction X, the distance between the light-blocking structure **30** corresponding to the light-emitting element **20** and the light-emitting element **20** may be smaller. Since the area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** is large, the lifespan of the light-emitting material of the first light-emitting element **201** may be longer, and the function undertaken by the area where the first light-emitting element **201** is located also may have a greater impact on the display effect. The brightness distribution of the first light-emitting element **201** may be relatively stronger in comparison with the second light-emitting element **202** with a smaller area. Therefore, in the present embodiment, in the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. That is, the first light-blocking structure **301** may be closer to the edge of the corresponding first light-emitting element **201**, and may block more light-emitting range of the corresponding first light-emitting element **201**. The distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be larger. That is, the second light-blocking structure **302** may be further away from the edge position of the corresponding second light-emitting element **202**, and may block less light-emitting range of the second light-emitting element **202**. Therefore, even the area S1 of the first light-emitting element **201** is larger than the area S2 of the second light-emitting element **202**, the proportions of the light-emitting elements **20** with different areas that are blocked by the light-blocking structure **30** at a large viewing angle may remain similar, balancing the difference of the white light characteristics of the light-emitting elements **20** with different areas at large viewing angles. Correspondingly, the uniformity of the white light characteristics of the display panel **000** at different viewing angles may be improved, to improve the display quality.

(49) In some embodiments shown in FIG. 5 which is another planar structure of the display panel, FIG. 6 which is an enlarged view of the M2 region in FIG. 5, and FIG. 7 which is an enlarged view of the M3 region in FIG. 5, the display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be lower than the transmittance of the second display region AA2.

(50) The colors of the first light-emitting elements **201** and the second light-emitting elements **202**

may be different (different colors are distinguished by different filling patterns in the figure). The first light-emitting elements **201** and the second light-emitting elements **202** may be all located in the first display region AA1. In another embodiment, the first light-emitting elements **201** and the second light-emitting elements **202** may be all located in the second display region AA2.

(51) In existing technologies, with the continuous development of display technology, a full screen using a camera under panel (CUP) area has become a new development trend. To ensure the photographing effect of the camera under panel area (CUP area), it is necessary to increase its transmittance to meet the amount of incoming light of photographing. Therefore, in the present disclosure, the display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be smaller than the transmittance of the second display region AA2, that is, the light transmittance of the second display region AA2 may be relatively high. Optionally, the first display region AA1 may be arranged at least partially around the second display region AA2. Optionally, while the second display region AA2 is used as a display region, it may also be used as a setting area corresponding to a photosensitive element, that is, the second display region AA2 may be used as the camera under panel area (CUP area) because of its high light transmittance. When the second display region AA2 is used as the setting area of the photosensitive element, it may need to have a higher transmittance, and correspondingly the higher transmittance may be achieved by setting the area of the light-transmitting area of the second display region AA2 to be larger than that of the first display region AA1. In some other embodiments, when the display panel **000** does not use the photosensitive element, the plurality of sub-pixels in the first display region AA1 and the second display region AA2 may display the screen normally together. When it is necessary to use the photosensitive element, for example, when it is necessary to use the camera for photographing, the larger light-transmitting area of the second display region AA2 may enable external light to enter the camera for imaging and shooting.

(52) For a display panel with a camera under panel area, whether in the first display region AA1 used as the normal display region or the second display region AA2 used as the camera under panel area, the different lifespan of the luminescent materials of different light-emitting elements on the display panel, the functions undertaken by the regions where different light-emitting elements are located, and the light-emitting colors of different light-emitting elements may have different effects on the display effect. Therefore, the display panels in the first display region AA1 may have different white light characteristics at a specific viewing angle, which easily affects the uniformity of the white light characteristics of the display panel in the first display region AA1 under different viewing angles and the display quality of the display panel. Therefore, how to improve the brightness distribution difference of white light in the first display region AA1 under a large viewing angle to improve the display uniformity of the entire first display region AA1, and how to improve the brightness distribution difference of white light in the second display region AA2 under a large viewing angle to improve the display uniformity in the entire second display region AA2, such that the display panel as a whole has better display uniformity, are problems to be solved.

(53) As shown in FIG. 5 and FIG. 6, in one embodiment, the first light-emitting elements **201** and the second light-emitting elements **202** may all be located in the first display region AA1, and the colors of the first light-emitting elements **201** and the second light-emitting elements **202** may be different. That is, the first light-emitting elements **201** and the second light-emitting elements **202** may all be light-emitting elements **20** with different colors in the first display region AA1 of the normal display region. The colors of the first light-emitting elements **201** and the second light-emitting elements **202** are different, and the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**. For example, the light-emitting color of the first light-emitting elements **201** may be blue or green, and the light-emitting

color of the second light-emitting elements **202** may be red. Along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. That is, the first light-blocking structure **301** may be closer to the edge of the corresponding first light-emitting element **201**, and may block more light-emitting range of the corresponding first light-emitting element **201** with a larger area. The distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be larger. That is, the second light-blocking structure **302** may be further away from the edge position of the corresponding second light-emitting element **202**, and may block less light-emitting range of the second light-emitting element **202** with a smaller area. Therefore, even the area S1 of the first light-emitting element **201** is larger than the area S2 of the second light-emitting element **202** which is located in the first display region AA1 together with the first light-emitting element **201**, the proportions of the light-emitting elements **20** with different areas that are blocked by the light-blocking structure **30** at a large viewing angle in the first display region AA1 may remain similar, balancing the difference of the white light characteristics of the light-emitting elements **20** with different areas at large viewing angles in the first display region AA1. Correspondingly, the uniformity of the white light characteristics of the first display region AA1 in the display panel **000** at different viewing angles may be improved, to improve the display quality.

(54) For description purposes only, the above embodiments where the light-emitting color of the first light-emitting elements **201** may be blue or green and the light-emitting color of the second light-emitting elements **202** may be red are used as examples to illustrate the present disclosure, and do not limit the scope of the present disclosure. In some other embodiments, the light-emitting color of the first light-emitting elements **201** may be red, and the light-emitting color of the second light-emitting elements **202** may be blue or green. The present disclosure has no limit on this.

(55) In another embodiment, as shown in FIG. 5 and FIG. 6, the first light-emitting elements **201** and the second light-emitting elements **202** may all be located in the second display region AA2, and the colors of the first light-emitting elements **201** and the second light-emitting elements **202** may be different. That is, the first light-emitting elements **201** and the second light-emitting elements **202** may all be light-emitting elements **20** with different colors in the second display region AA2 of the normal display region. The colors of the first light-emitting elements **201** and the second light-emitting elements **202** are different, and the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**. For example, the light-emitting color of the first light-emitting elements **201** may be blue or green, and the light-emitting color of the second light-emitting elements **202** may be red. Along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. That is, the first light-blocking structure **301** may be closer to the edge of the corresponding first light-emitting element **201**, and may block more light-emitting range of the corresponding first light-emitting element **201** with a larger area. The distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be larger. That is, the second light-blocking structure **302** may be further away from the edge position of the corresponding second light-emitting element **202**, and may block less light-emitting range of the second light-emitting element **202** with a smaller area. Therefore, even the area S1 of the first light-emitting element **201** is larger than the area S2 of the second light-emitting element **202** which is located in the second display region AA2 together with the first light-emitting element **201**, the proportions of the light-emitting elements **20** with different areas that are blocked by the light-blocking structure **30** at a large viewing angle in the second display region AA2 may remain similar, balancing the difference of the white light characteristics

of the light-emitting elements **20** with different areas at large viewing angles in the second display region AA2. Correspondingly, the uniformity of the white light characteristics of the second display region AA2 in the display panel **000** at different viewing angles may be improved, to improve the display quality.

(56) In some embodiments shown in FIG. 5, FIG. 8 which is an enlarged view of the M4 region in FIG. 5, and FIG. 9 which is another enlarged view of the M4 region in FIG. 5, the first light-emitting elements **201** may be located in the first display region AA1, and the second light-emitting elements **202** may be located in the second display region AA2, and the colors of the first light-emitting elements **201** and the second light-emitting elements **202** may be different. That is, the first light-emitting elements **201** and the second light-emitting elements **202** may be light-emitting elements **20** with different colors in the first display region AA1 of the normal display region and the second display region AA2 of the camera under panel area respectively. In one embodiment shown in FIG. 8, the light-emitting color of the first light-emitting elements **201** may be blue or green in the first display region AA1, and the light-emitting color of the second light-emitting elements **202** may be red in the second display region AA2. In another embodiment shown in FIG. 9, the light-emitting color of the first light-emitting elements **201** may be red in the first display region AA1, and the light-emitting color of the second light-emitting elements **202** may be blue or green in the second display region AA2. To ensure that the transmittance of the second display region AA2 is larger than the transmittance of the first display region AA1, the area of one light-emitting element **20** in the first display region AA1 may be larger than the area of one light-emitting element **20** in the second display region AA2 with a same color. As shown in FIG. 8, in one embodiment, when the light-emitting color of the first light-emitting elements **201** may be blue in the first display region AA1 and the light-emitting color of the second light-emitting elements **202** may be red in the second display region AA2, the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**, and along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. As shown in FIG. 9, in another embodiment, when the light-emitting color of the first light-emitting elements **201** may be red in the first display region AA1 and the light-emitting color of the second light-emitting elements **202** may be blue in the second display region AA2, the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be slightly smaller than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**, and along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. That is, the first light-blocking structure **301** may be closer to the edge of the corresponding first light-emitting element **201**, and may block more light-emitting range of the corresponding first light-emitting element **201** with a relatively smaller area. The transmittance of the second display region AA2 may be ensured to be high, to improve the photosensitive effect of the second display region AA2 when being used as the camera under panel area.

(57) In some embodiments shown in FIG. 5 and FIG. 10 which is another enlarged view of the M4 region in FIG. 5, the display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be lower than the transmittance of the second display region AA2.

(58) The colors of the first light-emitting elements **201** and the second light-emitting elements **202** may be same (same colors are represented by same filling patterns in the figure). The first light-emitting elements **201** may be located in the first display region AA1, and the second light-emitting elements **202** may be located in the second display region AA2.

(59) The display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be smaller than the transmittance of the second display region AA2, that is, the light transmittance of the second display region AA2 may be relatively high. Optionally, the first display region AA1 may be arranged at least partially around the second display region AA2. Optionally, while the second display region AA2 is used as a display region, it may also be used as a setting area corresponding to a photosensitive element, that is, the second display region AA2 may be used as the camera under panel area (CUP area) because of its high light transmittance. When the second display region AA2 is used as the setting area of the photosensitive element, it may need to have a higher transmittance, and correspondingly the higher transmittance may be achieved by setting the area of the light-transmitting area of the second display region AA2 to be larger than that of the first display region AA1. In some other embodiments, when the display panel **000** does not use the photosensitive element, the plurality of sub-pixels in the first display region AA1 and the second display region AA2 may display the screen normally together. When it is necessary to use the photosensitive element, for example, when it is necessary to use the camera for photographing, the larger light-transmitting area of the second display region AA2 may enable external light to enter the camera for imaging and shooting.

(60) In some embodiments shown in FIG. 5 and FIG. 10 which is another enlarged view of the M4 region in FIG. 5, the first light-emitting elements **201** may be located in the first display region AA1, and the second light-emitting elements **202** may be located in the second display region AA2, and the colors of the first light-emitting elements **201** and the second light-emitting elements **202** may be same. That is, the first light-emitting elements **201** and the second light-emitting elements **202** may be light-emitting elements **20** with same colors in the first display region AA1 of the normal display region and the second display region AA2 of the camera under panel area respectively. In one embodiment shown in FIG. 8, the light-emitting color of the first light-emitting elements **201** may be any one of red, blue or green in the first display region AA1, and the light-emitting color of the second light-emitting elements **202** may be any one of red, blue or green in the second display region AA2 but same as the first light-emitting elements **201**. To ensure that the transmittance of the second display region AA2 is larger than the transmittance of the first display region AA1, for the first light-emitting elements **201** and the second light-emitting elements **202** with same colors, the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**. Different areas of the different light-emitting elements may have different effects on the display effect. Therefore, the light-emitting elements with same colors in the first display region AA1 and in the second display region AA2 may have different white light characteristics at a specific viewing angle, which easily affects the uniformity of the white light characteristics of the display panel in the first display region AA1 under different viewing angles and the display quality of the display panel. In one embodiment, the area S1 of the orthographic projection of one first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of one second light-emitting element **202** on the substrate **10**, and along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be smaller than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. That is, the first light-blocking structure **301** in the first display region AA1 may be closer to the edge of the corresponding first light-emitting element **201**, and may block more light-emitting range of the corresponding first light-emitting element **201** with a larger area. The distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be larger. That is, the second light-blocking structure **302** in the second display region AA2 may be further away from the edge position of the corresponding second light-emitting element **202**, and may block less light-emitting range of the second light-emitting element **202** with a smaller area. Therefore, even the areas of the first light-emitting element **201** and the second light-

emitting element **202** with same colors are different, the proportions of the light-emitting elements **20** with different areas but same colors that are blocked by the light-blocking structure **30** at a large viewing angle in the first display region AA1 and the second display region AA2 may remain similar, balancing the difference of the white light characteristics of the light-emitting elements **20** with different areas at large viewing angles in the second display region AA2. Correspondingly, the uniformity of the white light characteristics of the second display region AA2 in the display panel **000** at different viewing angles may be improved, to improve the display quality.

(61) In some embodiments shown in FIG. 5, FIG. 11 which is an enlarged view of the M5 region in FIG. 5, and FIG. 12 which is another enlarged view of the M4 region in FIG. 5, the display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be lower than the transmittance of the second display region AA2.

(62) The second display region AA2 may at least include a first region AA21 and a AA22 which at least partially surrounds the first region AA21. The transmittance of the first region AA21 may be larger than the transmittance of the second region AA22.

(63) In the present embodiment, the display panel **000** may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be smaller than the transmittance of the second display region AA2, that is, the light transmittance of the second display region AA2 may be relatively high. Optionally, the first display region AA1 may be arranged at least partially around the second display region AA2. Optionally, while the second display region AA2 is used as a display region, it may also be used as a setting area corresponding to a photosensitive element, that is, the second display region AA2 may be used as the camera under panel area (CUP area) because of its high light transmittance. When the second display region AA2 is used as the setting area of the photosensitive element, it may need to have a higher transmittance, and correspondingly the higher transmittance may be achieved by setting the area of the light-transmitting area of the second display region AA2 to be larger than that of the first display region AA1. In some other embodiments, when the display panel **000** does not use the photosensitive element, the plurality of sub-pixels in the first display region AA1 and the second display region AA2 may display the screen normally together. When it is necessary to use the photosensitive element, for example, when it is necessary to use the camera for photographing, the larger light-transmitting area of the second display region AA2 may enable external light to enter the camera for imaging and shooting.

(64) In the present embodiment, to ensure the high transmittance of the second display region AA2, the transmittance of different regions in the second display region AA2 may be designed to be different. Specifically, the second display region AA2 may at least include a first region AA21 and a AA22 which at least partially surrounds the first region AA21. The transmittance of the first region AA21 may be larger than the transmittance of the second region AA22. In the actual design of the display panel **000**, it is desirable that the transmittance of the second display region AA2 used as the camera under panel area be as high as possible. To improve the difference in white light characteristics of the second display region AA2 under a large viewing angle, the second display region AA2 may also need to be provided with a light-blocking structure **30**. When the user observes the display panel, the white light difference at a specific viewing angle perceived by the user may be easy to observe under the large viewing angle of the display panel **000** and may not be perceived when viewed directly above the display panel **000**. In the present embodiment, the second display region AA2 may include at least the first area AA21 and the second area AA22 arranged at least partially around the first area AA21. The transmittance of the first area AA21 may be larger than that of the second area AA22. That is, the transmittance of the first area AA21 near the middle area of the second display region AA2 may be set to be larger than the transmittance of the second area AA22 in the periphery of the second display region AA2, to further improve the overall transmittance of the second display region AA2 by increasing the transmittance of the first area AA21.

(65) Optionally, as shown in FIG. 11, in the first area AA21, the light-blocking structures 30 may not be provided between at least some of the adjacent light-emitting elements 20, that is, the light-blocking structures 30 may be disposed between adjacent light-emitting elements 20 only in the second area AA22 at the periphery of the second display region AA2. Correspondingly, the overall transmittance of the second display region AA2 may be further improved by increasing the transmittance of the first area AA21, and the difference in white light characteristics under a large viewing angle of the second display region AA2, that is, when viewed obliquely, may be improved, too.

(66) Optionally, as shown in FIG. 12, the width of the light-blocking structures 30 in the first area AA21 in the first direction X may be set as W01, and the width of the light-blocking structures 30 in the second area AA22 in the first direction X may be set as W02, where $W01 < W02$. That is, both the first area AA21 and the second area AA22 in the second display region AA2 may be provided with the light-blocking structures 30, but the width W01 of the light-blocking structures 30 in the first area AA21 in the first direction X may be smaller than the width W02 of the light-blocking structures 30 in the second area AA22 in the first direction X. The width of the light-blocking structures 30 in the second area AA22 in the first direction X may be narrow. Correspondingly, the overall transmittance of the second display region AA2 may be further improved by increasing the transmittance of the first area AA21, and the difference in white light characteristics under a large viewing angle of the second display region AA2, that is, when viewed obliquely, may be improved, too.

(67) The embodiments shown in FIG. 11 and FIG. 12 with the arrangement structure of the light-emitting elements 20 of different colors in the second display region AA2 in the display panel 000 are used as examples only, for illustrating that the width W01 of the light-blocking structures 30 in the first area AA21 in the first direction X may be smaller than the width W02 of the light-blocking structures 30 in the second area AA22 in the first direction X. In various embodiments, the arrangement structure of the light-emitting elements 20 of different colors in the second display region AA2 may include, but is not limited to, the above structures, but may be other structures. It only needs to satisfy that in the light-emitting elements 20 with different areas in the first direction X, the distances between the light-emitting elements 20 and their corresponding light-blocking structures 30 may be different, and the width W01 of the light-blocking structures 30 in the first area AA21 in the first direction X may be smaller than the width W02 of the light-blocking structures 30 in the second area AA22 in the first direction X, to ensure the high transmittance of the second display region AA2. The present disclosure has no limit on this.

(68) In some embodiments shown in FIG. 5, FIG. 13 which is an enlarged view of the M4 region in FIG. 5, and FIG. 14 which is a cross-sectional view along the B-B' direction in FIG. 13, the display panel 000 may include a first display region AA1 and a second display region AA2. The transmittance of the first display region AA1 may be lower than the transmittance of the second display region AA2.

(69) The display panel 000 may further include a light-blocking layer 300.

(70) In the first display region AA1, the light-blocking layer 300 may include first hollow regions 300A. An orthographic projection of one first hollow region 300A on the substrate 10 may overlap one corresponding light-emitting element 20 in the first display region AA1.

(71) The first light-emitting elements 201 and the second light-emitting elements 202 may be located in the second display region AA2. The first light-blocking structures 301 and the second light-blocking structures 302 may be independent from each other.

(72) In the present embodiment, the display panel 000 may include a first display region AA1 and a second display region AA2. Optionally, the first display region AA1 may be arranged at least partially around the second display region AA2. Optionally, while the second display region AA2 is used as a display region, it may also be used as a setting area corresponding to a photosensitive element, that is, the second display region AA2 may be used as the camera under panel area (CUP

area). The transmittance of the first display region AA1 may be smaller than the transmittance of the second display region AA2, that is, the light transmittance of the second display region AA2 may be relatively high. In some other embodiments, when the display panel **000** does not use the photosensitive element, the plurality of sub-pixels in the first display region AA1 and the second display region AA2 may display the screen normally together. When it is necessary to use the photosensitive element, for example, when it is necessary to use the camera for photographing, the larger light-transmitting area of the second display region AA2 may enable external light to enter the camera for imaging and shooting.

(73) In the present embodiment, to ensure the high transmittance of the second display region AA2, the display panel **000** may be provided with a light-blocking layer **300**. The light-blocking layer **300** may be disposed on a side of the cathode layer **70** in the display panel **000** away from the substrate **10**, or the light-blocking layer **300** may also be disposed at other film positions on the side of the light-emitting material layers of the light-emitting elements **20** away from the substrate **10**. The present disclosure does not specifically limit the position of the film layer where the light-blocking layer **300** is provided. In the first display region AA1, the light-blocking layer **300** includes first hollow areas **300A**, and the orthographic projection of one first hollow area **300A** on the substrate **10** may overlap one corresponding light-emitting element **20** in the first display region AA1. That is, one position where one first hollow area **300A** is set in the first display region AA1 may be used for arranging one corresponding light-emitting elements **20** in the first display region AA1, such that one light-emitting element **20** in the first display region AA1 is able to emit light through the corresponding first hollow area **300A**. The light-blocking layers **300** in the remaining positions of the first display region AA1 may be connected together as an integral structure for blocking light crosstalk between adjacent light-emitting elements **20**. In the second display region AA2, the first light-emitting elements **201** and the second light-emitting elements **202** may be all located in the second display region AA2, and the first light-blocking structures **301** corresponding to the first light-emitting elements **201** and the second light-blocking structures **302** corresponding to the second light-emitting element **202** may be independent of each other. That is, the light-blocking structures **30** corresponding to the light-emitting elements **20** in the second display region AA2 may be independent structures, and may be not structurally connected to each other. Therefore, light-transmitting areas may be existed between different light-blocking structures **30** in the second display region AA2, which is beneficial to further improve the transmittance of the second display region AA2 of the display panel **000** and ensure the photosensitive effect when the photosensitive element is arranged in the second display region AA2.

(74) In one embodiment, as shown in FIG. **14**, the first light-blocking structures **301** and the second light-blocking structures **302** in the second display region AA2 may be disposed in the light-blocking layer **300**, that is, the light-blocking structures between the light-emitting elements **20** in the first display region AA1 and the light-blocking structures in the second display region AA2 may be formed in the same layer using a same process. In some other embodiments, the first light-blocking structures **301** and the second light-blocking structures **302** in the second display region AA2 may be disposed by providing another light-blocking layer at a side of the light-emitting material layers of the light-emitting elements **20** away from the substrate **10**. The present disclosure has no limit on this.

(75) In some embodiments shown in FIG. **1** to FIG. **3**, $|S1-S2| > |d1-d2|$.

(76) In some embodiments, in the display panel **000**, the area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be different from the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10**, and the area of the orthographic projection of the first opening area **600A** in the pixel definition layer **60** corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202** on the substrate **10**. Therefore, along the

first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding light-emitting element **201** may be also different from the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**. The light emission of the sub-pixel corresponding to each light-emitting element **20** may mainly depend on the light-emitting area of the light-emitting element **20**, that is, the area of the orthographic projection of one light-emitting element **20** on the substrate **10** may have large influence on the brightness and the color coordinates at a specific viewing angle of the sub-pixel where it is located. The relative brightness of the sub-pixel corresponding to each light-emitting element **20** after being blocked by the corresponding light-blocking structure **30** may have relatively small change. That is, compared with the influence of the light-blocking structures **30** on different light-emitting elements **20**, the area of the orthographic projection of each light-emitting element **20** itself on the substrate **10** may have a larger impact on the light-emitting brightness. Therefore, in the present embodiment, along the direction parallel to the plane of the display panel **000**, the change of the distances between two light-emitting elements **20** and their corresponding light-blocking structures **30** may be larger than the area change of the two light-emitting elements **20**, that is, $|S1-S2| > |d1-d2|$. Optionally, when $S1 > S2$, $d1 < d2$, it may be $S1-S2 > d2-d1$; or when $S1 > S2$, $d1 > d2$, it may be $S1-S2 > d1-d2$. Correspondingly, according to the difference between the influence on the brightness and the color coordinates at a specific viewing angle of the sub-pixel where the light-emitting element **20** is located by the area of the light-emitting element **20** and the influence on the relative brightness by the light-blocking structure **30** blocking the corresponding light-emitting element **20**, the difference in white light characteristics of different light-emitting elements **20** at a specific viewing angle may be adjusted mainly by changing the areas of the light-emitting elements **20**. This configuration combined with the different configuration of the light-blocking structures may be used to make the distances between light-emitting elements **20** with different areas and their corresponding light-blocking structures **30** different, and adjust the difference in white light characteristics of different light-emitting elements **20** at specific viewing angles, such that the relative brightness of two light-emitting elements **20** with different areas after being blocked by the corresponding light-blocking structures **30** may be as consistent as possible. The viewing angle difference of the white light of the display panel **000** and the uniformity of the white light characteristics of the display panel **000** under different viewing angles may be improved, to improve the display quality of the display panel **000**.

(77) In some embodiments shown in FIG. 1, FIG. 15 which is another enlarged view of the M1 region in FIG. 1, and FIG. 16 which is a cross-sectional view along the C-C' direction in FIG. 15, the light-emitting elements may further include third light-emitting elements **203**. An area of an orthographic projection of one third light-emitting element **203** on the substrate **10** may be S3.

(78) The light-blocking structures **30** may further include third light-blocking structures **303**. An orthographic projection of one third light-blocking structure **303** may at least partially surround one corresponding third light-emitting element **203**. Along the first direction X, a distance from one third light-blocking structure **303** to one corresponding third light-emitting element **203** may be d3, where $S1 > S3$, $d1 < d3$, and/or $S2 > S3$, $d2 < d3$.

(79) One light-emitting element **20** may include one first light-emitting element **201**, one second light-emitting element **202**, and one third light-emitting element **203**. The first light-emitting element **201**, the second light-emitting element **202**, and the third light-emitting element **203** may be light-emitting elements in the display panel **000** with different colors. The area of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be S1, the area of the orthographic projection of the second light-emitting element **201** on the substrate **10** may be S2, and the area of an orthographic projection of the third light-emitting element **203** on the substrate **10** may be S3. The area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be understood as an area of an orthographic projection of one first opening area **600A** in the pixel definition layer **60** corresponding to the first light-emitting

element **201**, the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10** may be understood as an area of an orthographic projection of one second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202**, and the area S3 of the orthographic projection of the third light-emitting element **203** on the substrate **10** may be understood as an area of an orthographic projection of a third opening area **600C** in the pixel definition layer **60** corresponding to the third light-emitting element **203**.

(80) One opening area **600** of the pixel definition layer **60** may be formed by opening an opening **60K** in the pixel definition layer **60** penetrating through the thickness of the pixel definition layer **60** itself. The cross-sectional shape of the opening **60K** may be set according to actual needs. As shown in FIG. **16**, in one embodiment, the cross-section may be an inverted trapezoid, and in this case, the range of the opening region **600** of the pixel definition layer **60** may be based on the bottom edge of the cross-section of the opening **60K** whose shape is an inverted trapezoid.

(81) The display panel **000** may further include the light-blocking structures **30** at the side of the light-emitting elements **20** away from the substrate **10**. The light-blocking structures **30** may be disposed between different light-emitting elements **20** for blocking, to avoid light crosstalk between different light-emitting elements **20**. Optionally, the light-blocking structures **30** may be located on the side of the cathode layer **70** away from the substrate **10**. In some other optional embodiments, the light-blocking structures **30** may also be located in other film layers on the side of the light-emitting elements **20** away from the substrate **10**. The present disclosure has no limit on this.

(82) One light-blocking structure **30** may include one first light-blocking structure **301**, one second light-blocking structure **302** and one third light-blocking structure **303**. In one light-blocking structure **30**, the first light-blocking structure **301** may correspond to one first light-emitting element **201**. The orthographic projection of the first light-blocking structure **301** may at least partially surround the corresponding first light-emitting element **201** (not shown in the figure). In one embodiment, the orthographic projection of the first light-blocking structure **301** may surround the corresponding first light-emitting element **201**, as shown in FIG. **15**. In one light-blocking structure **30**, the second light-blocking structure **302** may correspond to one second light-emitting element **202**. The orthographic projection of the second light-blocking structure **302** may at least partially surround the corresponding second light-emitting element **202** (not shown in the figure). In one embodiment, the orthographic projection of the second light-blocking structure **302** may surround the corresponding second light-emitting element **202**, as shown in FIG. **15**. In one light-blocking structure **30**, the third light-blocking structure **303** may correspond to one third light-emitting element **203**. The orthographic projection of the third light-blocking structure **303** may at least partially surround the corresponding third light-emitting element **203** (not shown in the figure). In one embodiment, the orthographic projection of the third light-blocking structure **303** may surround the corresponding third light-emitting element **203**, as shown in FIG. **15**. That is, the first light-blocking structure **301** may be at least provided at a portion of the periphery of the first light-emitting element **201**, the second light-blocking structure **302** may be at least provided at a portion of the periphery of the second light-emitting element **202**, and the third light-blocking structure **303** may be at least provided at a portion of the periphery of the third light-emitting element **203**. It can be understood that the present disclosure does not specifically limit the material of the light-blocking structures **30**, and the material only needs to be opaque and be able to play a light-blocking role.

(83) With the development of display technology, the lifespan of the light-emitting materials of different light-emitting elements on the display panel, the functions undertaken by the regions where different light-emitting elements are located, and the light-emitting colors of different light-emitting elements have different effects on the display effect. The white light characteristics of the panels at specific viewing angles are different, which in turn easily affects the uniformity of the white light characteristics of the display panel at different viewing angles and affects the display

quality of the display panel. To solve the above problems, the display panel provided by the present disclosure may adjust different light-emitting elements to different degrees according to the display differences of different light-emitting elements, to ensure that the final display effect of the display panel is uniform and good. In existing technologies, the area of a general light-emitting element, that is, the area of the corresponding opening area of the pixel definition layer in the present disclosure, and the design of the light-blocking structures, are two factors that may significantly affect the values of relative brightness of each sub-pixel after being shielded by one corresponding light-blocking structure, the change of brightness and color coordinate of each sub-pixel under a specific viewing angle.

(84) In the display panel **000** provided by the present disclosure, along the first direction X, a distance from the first light-blocking structure **301** to the corresponding first light-emitting element **201** may be d_1 . A distance from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be d_2 , where $S_1 > S_2$ and $d_1 < d_2$. Correspondingly, along the first direction X, a distance from the third light-blocking structure **303** to the corresponding third light-emitting element **203** may be d_3 , where $S_1 > S_3$ and $d_1 < d_3$, and/or $S_2 > S_3$ and $d_2 < d_3$. That is, the area S_1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S_2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance d_1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d_2 between the second light-blocking structure **302** and the second light-emitting element **202**. Therefore, the area S_1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S_3 of the orthographic projection of the third light-blocking structure **303** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction X, the distance d_1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d_3 between the third light-blocking structure **303** and the third light-emitting element **203**, as shown in FIG. 15 and FIG. 16.

(85) In another embodiment, the area S_1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S_2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance d_1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d_2 between the second light-blocking structure **302** and the second light-emitting element **202**. Therefore, the area S_2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10** may be different from the area S_3 of the orthographic projection of the third light-blocking structure **303** on the substrate **10**. The area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10** may be different from the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction X, the distance

d2 between the second light-blocking structure **302** and the first light-emitting element **201** may be different from the distance d3 between the third light-blocking structure **303** and the third light-emitting element **203** (not shown in the figure).

(86) In another embodiment, the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction X, the distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d2 between the second light-blocking structure **302** and the second light-emitting element **202**. Therefore, the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area S3 of the orthographic projection of the third light-blocking structure **303** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction X, the distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance d3 between the third light-blocking structure **303** and the third light-emitting element **203**. The area S2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10** may be different from the area S3 of the orthographic projection of the third light-blocking structure **303** on the substrate **10**. The area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10** may be different from the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction X, the distance d2 between the second light-blocking structure **302** and the first light-emitting element **201** may be different from the distance d3 between the third light-blocking structure **303** and the third light-emitting element **203** (not shown in the figure).

(87) In another embodiment shown in FIG. 15 and FIG. 16, the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be larger than the area S2 of the orthographic projection of the second light-blocking structure **302** on the substrate **10**, and the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be larger than the area S3 of the orthographic projection of the third light-blocking structure **303** on the substrate **10**. That is, the area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be larger than the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**, and larger than the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction X, the distance d1 between the first light-blocking structure **301** and the first light-emitting element **201** may be smaller than the distance d2 between the second light-blocking structure **302** and the second light-emitting element **202**, and smaller than the distance d3 between the third light-blocking structure **303** and the third light-emitting element **203**.

(88) In another embodiment shown in FIG. 1, FIG. 17 which is another enlarged view of the M1 region in FIG. 1, and FIG. 18 which is a cross-sectional view along the D-D' direction in FIG. 17, the area S1 of the orthographic projection of the first light-blocking structure **301** on the substrate

10 may be larger than the area **S2** of the orthographic projection of the second light-blocking structure **302** on the substrate **10**, and the area **S2** of the orthographic projection of the second light-blocking structure **302** on the substrate **10** may be larger than the area **S3** of the orthographic projection of the third light-blocking structure **303** on the substrate **10**, that is $S1 > S2 > S3$. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be larger than the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**, and the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10** may be larger than the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Along the first direction **X**, the distance **d1** between the first light-blocking structure **301** and the first light-emitting element **201** may be smaller than the distance **d2** between the second light-blocking structure **302** and the second light-emitting element **202**, and the distance **d2** between the second light-blocking structure **302** and the second light-emitting element **202** may be smaller than the distance **d3** between the third light-blocking structure **303** and the third light-emitting element **203**, that is $d1 < d2 < d3$.

(89) In the present disclosure, the area **S1** of the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be different from the area **S2** of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area **S3** of the orthographic projection of the third light-blocking structure **303** on the substrate **10** may be different from one of the area **S1** of the orthographic projection of the first light-blocking structure **301** on the substrate **10** and the area **S2** of the orthographic projection of the second light-blocking structure **302** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. The area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10** may be different from one of the area of the orthographic projection of the first opening area **600A** in the pixel definition layer corresponding to the first light-emitting element **201** on the substrate **10** and the area of the orthographic projection of the second opening area **600B** in the pixel definition layer corresponding to the second light-emitting element **202** on the substrate **10**. Along the first direction **X**, the distance **d1** between the first light-blocking structure **301** and the first light-emitting element **201** may be different from the distance **d2** between the second light-blocking structure **302** and the second light-emitting element **202**. The distance **d3** between the third light-blocking structure **303** and the third light-emitting element **203** may be different from one of the distance **d1** between the first light-blocking structure **301** and the first light-emitting element **201** and the distance **d2** between the second light-blocking structure **302** and the second light-emitting element **202**.

Therefore, for the light-emitting elements **20** with different areas, for example, for a light-emitting element **20** with a larger area, the distance between one light-emitting element **20** and its corresponding light-blocking structure **30** in the direction parallel to the plane of the display panel **000** may be designed to be larger or smaller. Correspondingly, although the areas of the light-emitting elements **20** in the display panel **000** are different, by adjusting the distance between one light-emitting element **20** and its corresponding light-blocking structure **30** in the direction parallel to the plane of the display panel **000**, the relative brightness of two light-emitting elements **20** with different areas after being blocked by corresponding light-blocking structures **30** respectively may be kept consistent as much as possible, and the light-emitting brightness at a specific viewing angle may be also as consistent as possible. The display difference between different light-emitting elements **20** may be reduced, to improve the view angle difference of the white light of the display

panel **000**. The uniformity of white light characteristics of the display panel **000** under different viewing angles may be improved, thereby helping to improve the display quality of the display panel **000**.

(90) In one embodiment, the first light-emitting element **201** may be a blue light-emitting element, the second light-emitting element **202** may be a green light-emitting element, and the third light-emitting element **203** may be a red light-emitting element. In some other embodiments, the first light-emitting element **201** may be a blue light-emitting element, the second light-emitting element **202** may be a red light-emitting element, and the third light-emitting element **203** may be a green light-emitting element. In some other optional embodiments, the first light-emitting element **201** may be a green light-emitting element, the second light-emitting element **202** may be a blue light-emitting element, and the third light-emitting element **203** may be a red light-emitting element. In some other optional embodiments, the first light-emitting element **201** may be a green light-emitting element, the second light-emitting element **202** may be a red light-emitting element, and the third light-emitting element **203** may be a blue light-emitting element. In some other optional embodiments, the first light-emitting element **201** may be a red light-emitting element, the second light-emitting element **202** may be a blue light-emitting element, and the third light-emitting element **203** may be green light-emitting elements. In some other optional embodiments, the first light-emitting element **201** may be a red light-emitting element, the second light-emitting element **202** may be a green light-emitting element, and the third light-emitting element **203** may be a blue light-emitting element. The present disclosure has no limit on this.

(91) In some embodiments shown in FIG. 1, FIG. 17, and FIG. 18, $|(S1-S2)-(S2-S3)| > |(d1-d2)-(d2-d3)|$.

(92) In some embodiments, in the display panel **000**, the area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be different from the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10**, and the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10** may be different from the area S3 of the orthographic projection of the third light-emitting element **203** on the substrate **10**. The area of the orthographic projection of the first opening area **600A** in the pixel definition layer **60** corresponding to the first light-emitting element **201** on the substrate **10** may be different from the area of the orthographic projection of the second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202** on the substrate **10**, and the area of the orthographic projection of the second opening area **600B** in the pixel definition layer **60** corresponding to the second light-emitting element **202** on the substrate **10** may be different from the area of the orthographic projection of the third opening area **600C** in the pixel definition layer corresponding to the third light-emitting element **203** on the substrate **10**. Therefore, along the first direction X, the distance d1 from the first light-blocking structure **301** to the corresponding light-emitting element **201** may be also different from the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202**, and the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** may be different from distance d3 between the third light-blocking structure **303** and the third light-emitting element **203**. The light emission of the sub-pixel corresponding to each light-emitting element **20** may mainly depend on the light-emitting area of the light-emitting element **20**, that is, the area of the orthographic projection of one light-emitting element **20** on the substrate **10** may have large influence on the brightness and the color coordinates at a specific viewing angle of the sub-pixel where it is located. The relative brightness of the sub-pixel corresponding to each light-emitting element **20** after being blocked by the corresponding light-blocking structure **30** may have relatively small change. That is, compared with the influence of the light-blocking structures **30** on different light-emitting elements **20**, the area of the orthographic projection of each light-emitting element **20** itself on the substrate **10** may have a larger impact on the light-emitting brightness. Therefore, in the present embodiment, along the direction parallel to

the plane of the display panel **000**, the change of the distances between two light-emitting elements **20** and their corresponding light-blocking structures **30** may be larger than the area change of the two light-emitting elements **20**, that is, $|(S1-S2)-(S2-S3)| > |(d1-d2)-(d2-d3)|$. Optionally, when $S1 > S2 > S3$ and $d1 < d2 < d3$, it may be $(S1-S2)-(S2-S3) > (d3-d2)-(d2-d1)$; or when $S1 > S2 > S3$ and $d1 > d2 > d3$, it may be $(S1-S2)-(S2-S3) > (d1-d2)-(d2-d3)$. Correspondingly, according to the difference between the influence on the brightness and the color coordinates at a specific viewing angle of the sub-pixel where the light-emitting element **20** is located by the area of the light-emitting element **20** and the influence on the relative brightness by the light-blocking structure **30** blocking the corresponding light-emitting element **20**, the difference in white light characteristics of different light-emitting elements **20** at a specific viewing angle may be adjusted mainly by changing the areas of the light-emitting elements **20**. This configuration combined with the different configuration of the light-blocking structures may be used to make the distances between light-emitting elements **20** with different areas and their corresponding light-blocking structures **30** different, and adjust the difference in white light characteristics of different light-emitting elements **20** at specific viewing angles, such that the relative brightness of two light-emitting elements **20** with different areas after being blocked by the corresponding light-blocking structures **30** may be as consistent as possible. The viewing angle difference of the white light of the display panel **000** and the uniformity of the white light characteristics of the display panel **000** under different viewing angles may be improved, to improve the display quality of the display panel **000**.

(93) In some embodiments shown in FIG. 1, FIG. 19 which is another enlarged view of the M1 region in FIG. 1, and FIG. 20 which is a cross-sectional view along the E-E' direction in FIG. 19, an orthographic projection of one first light-blocking structure **301** on the substrate **10** may surround one corresponding first light-emitting element **201**, and an area of an opening formed by the first light-blocking structure **301** may be S11. An orthographic projection of one second light-blocking structure **302** on the substrate **10** may surround one corresponding second light-emitting element **202**, and an area of an opening formed by the second light-blocking structure **302** may be S21, where $S11 > S21$ and $S11/S21 = S1/S2$.

(94) In the display panel **000**, one first light-blocking structure **301** is disposed at a side of one corresponding first light-emitting element **201** away from the substrate **10** and the orthographic projection of the first light-blocking structure **301** to block the light with a large viewing angle emitted by the first light-emitting element **201**. At this time, the area of the opening formed by the first light-blocking structure **301** may be S11 (it can also be understood as the actual light-emitting area S11 of the first light-emitting element **201**). One second light-blocking structure **302** may be disposed at a side of one corresponding second light-emitting element **202** away from the substrate **10** and the orthographic projection of the second light-blocking structure **302** on the substrate **10** may surround the corresponding second light-emitting element **202** to block the light with a large viewing angle emitted by the second light-emitting element **202**. At this time, the area of the opening formed by the second light-blocking structure **302** may be S21 (it can also be understood as the actual light-emitting area S21 of the second light-emitting element **202**). After the first light-emitting element **201** and the second light-emitting element **202** with different areas are shielded by the first light-blocking structure **301** and the second light-blocking structure **302** respectively, the opening area S11 formed by the first light-blocking structure **301** which is the actual light-emitting area of the first light-emitting element **201** may be larger than the opening area S21 formed by the second light-blocking structure **302** which is the actual light-emitting area S21 of the second light-emitting element **202**. Assuming that the first light-emitting element **201** and the second light-emitting element **202** are two light-emitting elements **20** of different colors and the area S1 of the first light-emitting element **201** is larger than the area S2 of the second light-emitting element **202**, when the first light-emitting element **201** and the second light-emitting element **202** are viewed from the same large viewing angle, the first light-emitting element **201** may be blocked by the first light-blocking structure **301** and the second light-emitting element **202** may be blocked by the

second light-blocking structure **302**. The relative brightness of the first light-emitting element **201** and the second light-emitting element **20** the two light-emitting elements **20** may be different after each is blocked by its corresponding light-blocking structure **30**. In the present disclosure, a ration between the actual light-emitting area S11 after the first light-emitting element **201** is blocked by the first light-blocking structure **301** and the actual light-emitting area S21 after the second light-emitting element **202** is blocked by the second blocking structure **302** may be configured to be as close as possible to the ratio between the area S1 of the first light-emitting element **201** and the area S2 of the second light-emitting element **202**, that is, $S11/S21=S1/S2$, such that the light-emitting brightness of the first light-emitting element **201** and the second light-emitting element **20** may be as same as possible when the first light-emitting element **201** and the second light-emitting element **202** are viewed from the same large viewing angle. Therefore, the viewing angle difference of the white light of the display panel **000** and the uniformity of the white light characteristics of the display panel **000** under different viewing angles may be improved.

(95) Since the opening area S11 formed by the first light-blocking structure **301** is larger than the opening area S21 formed by the second light-blocking structure **302** and it is expected that the ratio between the actual light-emitting area S11 of the first light-emitting element **201** after being shielded by the first light-blocking structure **301** and the actual light-emitting area S21 of the second light-emitting element **202** blocked by the second light-blocking structure **302** is equal to the ratio of the area S1 of the first light-emitting element **201** to the area S2 of the second light-emitting element **202** as much as possible, based on that that the opening area S11 formed by the first light-blocking structure **301** is larger than the opening area S21 formed by the second light-blocking structure **302**, the distance d1 between the first light-emitting element **201** and its corresponding first light-blocking structure **301** may need to be relatively small, and the distance d2 between the second light-emitting element **202** and the corresponding second light-blocking structure **302** may need to be relatively larger. That is, along the first direction X, the distances between different light-emitting elements **20** and their corresponding light-blocking structures **30** may be different, to balance the difference of the white light characteristics of the light-emitting elements **20** with different areas at large viewing angles in the display panel **000**.

(96) In some embodiments shown in FIG. 1, FIG. 19, and FIG. 20, along the first direction X, the width of the first light-blocking structure **301** may be W1 and the width of the second light-blocking structure **302** may be W2 where $W1>W2$.

(97) In the display panel **000**, the width of the light-blocking structures **30** corresponding to different light-emitting elements **20** may be configured differently. For example, the width W1 of the first light-blocking structure **301** corresponding to the first light-emitting element **201** may larger than the width W2 of the second light-blocking structure **302** corresponding to the second light-emitting element **202**. Since the area S1 of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be larger than the area S2 of the orthographic projection of the second light-emitting element **202** on the substrate **10** and the distance d1 from the first light-blocking structure **301** to the corresponding light-emitting element **201** may be larger than the distance d2 from the second light-blocking structure **302** to the corresponding second light-emitting element **202** along the first direction X, the actual light-emitting viewing angle $\alpha 2$ of the second light-emitting element **202** may be larger than the actual light-emitting viewing angle $\alpha 1$ of the first light-emitting element **201** and the width W2 of the second light-blocking structure **302** corresponding to the second light-emitting element **202** may be reduced. That is, the width W1 of the first light-blocking structure **301** corresponding to the first light-emitting element **201** may larger than the width W2 of the second light-blocking structure **302** corresponding to the second light-emitting element **202**, along the first direction X. While satisfying the blocking of the light-emitting elements **20** by the corresponding light-blocking structures **30**, the area ratio of the light-blocking structures **30** in the display panel **000** may be reduced as much as possible, such that the transmittance of the corresponding area may be improved when the camera under panel technology

is applied in the display panel to ensure the photosensitive effect of the camera under the panel.

(98) In some embodiments, the display panel **000** may further include the third light-emitting elements **203** and the area of the third light-emitting elements **203** may be S3. The light-blocking structures **30** may further include the third light-blocking structures **303**. The orthographic projection of one third light-blocking structure **303** on the substrate **10** may surround one corresponding third light-emitting element **203**. Along the first direction X, a distance from one third light-blocking structure **303** to the corresponding third light-emitting element **203** may be d3. When $S1 > S2 > S3$ and $d1 < d2 < d3$, the actual light-emitting viewing angle of the third light-emitting element **203** may be the largest, and the actual light-emitting viewing angle $\alpha1$ of the first light-emitting element **201** may be the smallest. Therefore, along the first direction X, the width of the third light-blocking structure **303** corresponding to the third light-emitting element **203** may be set to be the smallest and smaller than the width W2 of the second light-blocking structure **302** corresponding to the second light-emitting element **202**. While satisfying the blocking of the light-emitting elements **20** by the corresponding light-blocking structures **30**, the area ratio of the light-blocking structures **30** in the display panel **000** may be reduced as much as possible, such that the transmittance of the corresponding area may be improved when the camera under panel technology is applied in the display panel to ensure the photosensitive effect of the camera under the panel.

(99) In some embodiments shown in FIG. 1, FIG. 19 which is another enlarged view of the M1 region in FIG. 1, and FIG. 21 which is a cross-sectional view along the E-E' direction in FIG. 19, along a direction Z perpendicular to the plane where the substrate **10** is located, a distance between a side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be H1, and a distance between a side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202** may be H2, where $|H1 - H2| > 0$.

(100) As shown in FIG. 21, along the direction Z perpendicular to the plane where the substrate **10** is located, the distance H1 between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be understood as a distance between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and a side surface of the light-emitting material layer of the corresponding first light-emitting element **201**, and the distance H2 between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202** may be understood as a distance between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and a side surface of the light-emitting material layer of the corresponding second light-emitting element **202**.

(101) The actual light-emitting viewing angles of different light-emitting elements **20** may also be affected by the height of corresponding light-blocking structures **30**. The height of one light-blocking structure **30** may be understood as a distance between a side surface of the light-blocking structure **30** away from one corresponding light-emitting element **20** and the corresponding light-emitting element **20** along the direction Z perpendicular to the plane where the substrate **10** is located. As shown in FIG. 21, in the present disclosure, the distance H1 between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be configured to be larger than the distance H2 between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**, that is $|H1 - H2| > 0$. Therefore, the actual light-emitting viewing angles of the first light-emitting element **201** and the second light-emitting element **202** may be different after being blocked by the corresponding light-blocking structures. In one embodiment, the area S1

of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be larger than the area **S2** of the orthographic projection of the second light-emitting element **202** on the substrate **10**. Correspondingly, along the direction **Z** perpendicular to the plane where the substrate **10** is located, the distance **H1** between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be configured to be larger than the distance **H2** between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**, that is, $S1 > S2$ and $H1 > H2$. Therefore, when the area of one opening area **600** of the pixel definition layer **60** corresponding to one light-emitting element **20** is larger, the distance between the side surface of the corresponding light-blocking structure **30** away from the substrate **10** and the light-emitting element **20** may be larger. Correspondingly, the first light-blocking structure **301** may block more of the light-emitting viewing angle of the first light-emitting element **201**, thereby reducing the actual light-emitting viewing angle of the first light-emitting element **201** and ensuring the actual light-emitting viewing angle of the second light-emitting element **202** with a small area as wide as possible. The difference of the white light characteristics of the light-emitting elements **20** with different areas at large viewing angles in the display panel **000** may be balanced, to improve the uniformity of the white light characteristics at different view angles and the display quality of the display panel **000**.

(102) For description purposes only, the above embodiments where the area **S1** of the orthographic projection of the first light-emitting element **201** on the substrate **10** is larger than the area **S2** of the orthographic projection of the second light-emitting element **202** on the substrate **10** and the distance **H1** from the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** to the corresponding first light-emitting element **201** is larger than the distance **H2** from the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** to the corresponding second light-emitting element **202** (that is, $S1 > S2$ and $H1 > H2$), are used as examples to illustrate the present disclosure, and do not limit the scope of the present disclosure. In some other embodiment, the area **S1** of the orthographic projection of the first light-emitting element **201** on the substrate **10** may be larger than the area **S2** of the orthographic projection of the second light-emitting element **202** on the substrate **10**. Correspondingly, along the direction **Z** perpendicular to the plane where the substrate **10** is located, the distance **H1** between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be configured to be smaller than the distance **H2** between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**, that is, $S1 > S2$ and $H1 > H2$. The present disclosure has no limit on this as long as the height of the light-blocking structures **30** corresponding to different light-emitting elements **20** are different.

(103) In some embodiments, the first light-blocking structure **301** and the second light-blocking structure **302** may be formed in a same layer with a same material. Along the direction **Z** perpendicular to the plane where the substrate **10** is located, the distance **H1** between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be configured to be larger than the distance **H2** between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**. That may be understood as that a thickness **J1** of the first light-blocking structure **301** may be larger than a thickness **J2** of the second light-blocking structure **302** along the direction **Z** perpendicular to the plane where the substrate **10** is located. Correspondingly, the first light-blocking structure **301** may block more of the light-emitting viewing angle of the first light-emitting element **201**, thereby reducing the actual light-emitting viewing angle of the first light-

emitting element **201** and ensuring the actual light-emitting viewing angle of the second light-emitting element **202** with a small area as wide as possible.

(104) In some other embodiments shown in FIG. **1**, FIG. **19** which is another enlarged view of the M1 region in FIG. **1**, and FIG. **22** which is a cross-sectional view along the E-E' direction in FIG. **19**, the first light-blocking structure **301** and the second light-blocking structure **302** may be formed in different layers with a same material. Along the direction Z perpendicular to the plane where the substrate **10** is located, the distance H1 between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** may be configured to be larger than the distance H2 between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**. That may be understood as that a film layer where the first light-blocking structure **301** is located may be disposed at a side of a film layer where the second light-blocking structure **302** is located away from the substrate **10** and a thickness J1 of the first light-blocking structure **301** may be equal to a thickness J2 of the second light-blocking structure **302**. Correspondingly, the first light-blocking structure **301** may block more of the light-emitting viewing angle of the first light-emitting element **201**, thereby reducing the actual light-emitting viewing angle of the first light-emitting element **201** and ensuring the actual light-emitting viewing angle of the second light-emitting element **202** with a small area as wide as possible.

(105) In some other embodiments shown in FIG. **1**, FIG. **19**, FIG. **21**, FIG. **23** which is another enlarged view of the M1 region in FIG. **1**, and FIG. **24** which is a cross-sectional view along the F-F' direction in FIG. **23**, the height of the light-blocking structures **30** that is, the distances between the side surfaces of the light-blocking structures **30** away from the corresponding light-emitting elements **20** and the corresponding light-emitting element **20** along the direction Z perpendicular to the plane where the substrate **10** is located, may be configured differently. Therefore, for the first light-emitting element **21** with a larger area, it may be equal to increase the distance H1 between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** along the direction Z perpendicular to the plane where the substrate **10** is located. The light-emitting viewing angle of the first light-emitting element **201** after being blocked by the first light-blocking structure **301** may be reduced further. In one embodiment shown in FIG. **19** and FIG. **21**, the area of the first light-emitting element **201** may be S1 and the area of the second light-emitting element **202** may be S2, where $S1 > S2$. The distance between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X may be d1, and the distance between the second light-emitting element **202** and the corresponding second light-blocking structure **302** may be d2, where $d1 < d2$. Therefore, after in the direction Z perpendicular to the plane where the substrate **10** is located, after the distance H1 between the side surface **301E** of the first light-blocking structure **301** away from the corresponding first light-emitting element **201** and the corresponding first light-emitting element **201** is configured to be larger than the distance H2 between the side surface **302E** of the second light-blocking structure **302** away from the corresponding second light-emitting element **202** and the corresponding second light-emitting element **202**, the distance between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X may be appropriately increased, as shown in FIG. **23** and FIG. **24**. That is, the distance d1' between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X in FIG. **24** may be larger than the distance d1 between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X in FIG. **21**. Based on that the distance d2 between the second light-emitting element **202** and the corresponding second light-blocking structure **302** in the first direction X in FIG. **21** is same as the distance d2 between the second light-emitting element **202** and the corresponding second light-blocking structure **302** in the first direction X in FIG. **23**,

the difference between the distance $d1'$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X in FIG. **24** and the distance $d2$ between the second light-emitting element **202** and the corresponding second light-blocking structure **302** in the first direction X may still exist, but may be reduced. It may also be understood that $|d1-d2|$ in FIG. **21** is larger than $|d1'-d2|$ in FIG. **24**, to avoid the differential design of the distances between the light-blocking structures **30** corresponding different light-emitting elements **20** as much as possible. The process difficulty may be reduced and the process efficiency may be improved. Further, the distance $d1'$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X in FIG. **24** may be larger than that the distance $d1$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X in FIG. **21**. That may be equivalent to reducing the width of the first light-emitting element **201** to the corresponding first light-blocking structure **301** in the first direction X in FIG. **24**, and the width of the first light-blocking structure **301** in the first direction X may be smaller than the width of the first light-blocking structure **301** in the first direction X in FIG. **21**, which may further improve the transmittance of the display panel.

(106) In some embodiments shown in FIG. **1** to FIG. **3**, a distance between the first light-blocking structure **301** and the corresponding first light-emitting element **201** in the first direction X may be $d01$, and a distance between the first light-blocking structure **301** and the corresponding first light-emitting element **201** in a second direction Y may be $d02$, where $d01=d02$ and the second direction Y interests the first direction X in the direction parallel to the plane where the display panel **000** is located.

(107) For one light-emitting element **20**, for example, for one first light-emitting element **201**, the distance $d01$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X may be equal to the distance $d02$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the second direction Y. Optionally, the orthographic projection of the first light-emitting element **201** on the substrate **10** may be a circle, and the orthographic projection of the first light-blocking structure **301** on the substrate **10** may be an annulus. The circle and the annulus may be concentric structures such that the distance $d01$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the first direction X may be equal to the distance $d02$ between the first light-emitting element **201** and the corresponding first light-blocking structure **301** in the second direction Y. A distance between the second light-blocking structure **302** and the corresponding second light-emitting element **202** in the first direction X may be $d03$, and a distance between the second light-blocking structure **302** and the corresponding second light-emitting element **202** in the second direction Y may be $d04$, where $d03=d04$. The second direction Y may be any direction different from the first direction parallel to the plane where the display panel **000** is located. Correspondingly, when the first light-emitting element **201** is observed from different viewing angles, the brightness difference of the different viewing angles may be reduced, to improve the uniformity of the brightness distribution under different viewing angles and the display uniformity of the light-emitting elements **20**.

(108) In some embodiments shown in FIG. **1**, FIG. **2**, FIG. **5**, and FIG. **25** which is another enlarged view of the M1 region in FIG. **1**, the orthographic projection of one light-emitting element **20** on the substrate **10** may be anyone of a circle, an oval, or a chamfered rectangle.

(109) In the display panel **000**, the orthographic projection of one light-emitting element **20** on the substrate **10** may be anyone of a circle, an oval, or a chamfered rectangle, to prevent the shape of the orthographic projection of the light-emitting element **20** on the substrate **10** from having right-angled edges and make the orthographic projection of the light-emitting element **20** on the substrate **10** to be close to a circle as much as possible. Therefore, diffraction energy may be distributed evenly in different directions, and the diffraction energy in a single direction may be reduced. Especially when the display panel **000** includes the first display region AA1 and the second display

region AA2 shown in FIG. 5, the transmittance of the first display region AA1 may be smaller than the transmittance of the second display region AA2. The second display region AA2 may also be used as the setting area corresponding to the photosensitive element when being used as the display region. When the second display region AA2 is used as the setting area corresponding to the photosensitive element, at least the shape of the orthographic projection of the light-emitting element **20** in the second display region on the substrate **10** may be configured to be any one of a circle, an ellipse, or a chamfered rectangle. Therefore, the diffraction problem in the photosensitive element setting area may be effectively alleviated, to further improve optical effect of the camera under the panel in the second display region AA2.

(110) The embodiment in FIG. 2 where the orthographic projection of one light-emitting element **20** on the substrate **10** is a circle is used as an example to illustrate the present disclosure. For the same light-emitting element **20**, along the direction parallel to the plane of the display panel **000**, the distance from the corresponding light-blocking structure **30** to the light-emitting element **20** may need to be equal in all directions, to reduce the brightness difference under different viewing angles when the light-emitting element **20** is observed under different viewing angles and improve the uniformity of the brightness distribution under different viewing angles. To achieve this, the light-blocking structure **30** corresponding to the circular light-emitting element **20** may be an annular structure, and the orthographic projection of the light-blocking structure **30** of the annular structure on the substrate **10** may surround the circular light-emitting element **20**.

(111) In another embodiment shown in FIG. 25, the orthographic projection of one light-emitting element **20** on the substrate **10** may be a chamfered rectangle. The light-blocking structure **30** corresponding to the light-emitting element **20** of a chamfered rectangle may also be an annular structure. The annular light-blocking structure **30** may include an inner edge close to the light-emitting element **20** and an outer edge away from the light-emitting element **20**. At least the inner edge may be a chamfered rectangular structure matching the shape of the light-emitting element **20**. The orthographic projection of the light-blocking structure **30** of the annular structure on the substrate **10** may surround the circular light-emitting element **20**. Therefore, for one same light-emitting element **20**, along the direction parallel to the plane of the display panel **000**, the distance from the corresponding light-blocking structure **30** to the light-emitting element **20** may need to be equal in all directions, to reduce the brightness difference under different viewing angles when the light-emitting element **20** is observed under different viewing angles and improve the uniformity of the brightness distribution under different viewing angles.

(112) The above embodiments where the orthographic projection of one light-emitting element **20** on the substrate **10** is one of the above structures are used as examples to illustrate the present disclosure, and do not limit the scopes of the present disclosure. In various other embodiments, the orthographic projection of one light-emitting element **20** on the substrate **10** is not limited to above structures, and may be configured to be other shapes that could reduce the diffraction effect.

(113) The present disclosure also provides a display device. As shown in FIG. 26, in one embodiment, the display device **111** may include any display panel **000** provided by various embodiments of the present disclosure. The embodiment shown in FIG. 26 where the display device **111** is a cell phone is used as an example only to illustrate the present disclosure, and does not limit the scopes of the present disclosure. In some other embodiments, the display device **111** may be any display device with display function, such as a computer, a television, a touch controller, or a vehicle display device.

(114) In the present disclosure.

(115) Various embodiments have been described to illustrate the operation principles and exemplary implementations. It should be understood by those skilled in the art that the present disclosure is not limited to the specific embodiments described herein and that various other obvious changes, rearrangements, and substitutions will occur to those skilled in the art without departing from the scope of the disclosure. Thus, while the present disclosure has been described in

detail with reference to the above described embodiments, the present disclosure is not limited to the above described embodiments, but may be embodied in other equivalent forms without departing from the scope of the present disclosure, which is determined by the appended claims.

Claims

1. A display panel, comprising: a substrate; light-emitting elements on a side of the substrate; light-blocking structures on a side of the light-emitting elements away from the substrate, wherein: the light-emitting elements include first light-emitting elements and second light-emitting elements; an area of an orthographic projection of a first light-emitting element on the substrate is S1, and an area of an orthographic projection of a second light-emitting element on the substrate is S2; a light-blocking structure includes a first light-blocking structure and a second light-blocking structure; an orthographic projection of the first light-blocking structure on the substrate at least partially surrounds a corresponding first light-emitting element and a distance from the first light-blocking structure to the corresponding first light-emitting element along a first direction is d1; an orthographic projection of the second light-blocking structure on the substrate at least partially surrounds a corresponding second light-emitting element and a distance from the second light-blocking structure to the corresponding second light-emitting element along the first direction is d2, wherein the first direction is parallel to a plane where the display panel is located; and $|S1-S2|>0$, and $|d1-d2|>0$.
2. The display panel according to claim 1, wherein $S1>S2$ and $d1<d2$.
3. The display panel according to claim 2, wherein: the display panel includes a first display area and a second display area, wherein the transmittance of the first display area is smaller than the transmittance of the second display area; the first light-emitting elements and the second light-emitting elements have different colors; and the first light-emitting elements and the second light-emitting elements are located in the first display area; or the first light-emitting elements and the second light-emitting elements are located in the second display area.
4. The display panel according to claim 2, wherein: the display panel includes a first display area and a second display area, wherein the transmittance of the first display area is smaller than the transmittance of the second display area; the first light-emitting elements and the second light-emitting elements have same colors; and the first light-emitting elements are located in the first display area and the second light-emitting elements are located in the second display area.
5. The display panel according to claim 1, wherein: the display panel includes a first display area and a second display area, wherein the transmittance of the first display area is smaller than the transmittance of the second display area; and the second display area includes at least a first region and a second region at least partially surrounding the first region, wherein the transmittance of the first region is larger than the transmittance of the second region.
6. The display panel according to claim 5, wherein: the first light-blocking structure is not provided between two adjacent light-emitting elements of a portion of the light-emitting elements in the first region.
7. The display panel according to claim 5, wherein: a width of the light-blocking structure in the first region along the first direction is W01, and a width of the light-blocking structure in the second region along the first direction is W02, wherein $W01<W02$.
8. The display panel according to claim 1, wherein: the display panel includes a first display area and a second display area, wherein the transmittance of the first display area is smaller than the transmittance of the second display area; the display panel further includes a light-blocking layer; in the first display area, the light-blocking layer includes first hollow regions, wherein an orthographic projection of one first hollow region overlaps one corresponding light-emitting element in the first display region; and the first light-emitting elements and the second light-emitting elements are located in the second display area, and the first light-blocking structures and

the second light-blocking structures are independent from each other.

9. The display panel according to claim 1, wherein:

$$|S1-S2|>|d1-d2|.$$

10. The display panel according to claim 1, further including a third light-emitting element, wherein: an area of an orthographic projection of the third light-emitting element on the substrate is $S3$; the light-blocking structure further includes a third light-blocking structure; an orthographic projection of the third light-blocking structure on the substrate at least partially surrounds the third light-emitting element; and along the first direction, a distance from the third light-blocking structure to the third light-emitting element is $d3$, wherein, $S1>S3$, $d1<d3$, and/or, $S2>S3$, $d2<d3$.

11. The display panel according to claim 10, wherein:

$$S1>S2>S3, \text{ and } d1<d2<d3.$$

12. The display panel according to claim 11, wherein: $|(S1-S2)-(S2-S3)|>|(d1-d2)-(d2-d3)|$.

13. The display panel according to claim 2, wherein: the orthographic projection of one first light-blocking structure on the substrate surrounds one corresponding first light-emitting element; an area of an opening area formed by the first light-blocking structure is $S11$; the orthographic projection of one second light-blocking structure on the substrate surrounds one corresponding second light-emitting element; an area of an opening area formed by the second light-blocking structure is $S21$; and $S11>S21$, and $S11/S21=S1/S2$.

14. The display panel according to claim 2, wherein: along the first direction, a width of one first light-blocking structure is $W1$ and a width of one second light-blocking structure is $W2$, wherein $W1>W2$.

15. The display panel according to claim 1, wherein: along a direction perpendicular to the plane where the substrate is located, a distance from a side surface of one first light-blocking structure away from one corresponding first light-emitting element to the corresponding first light-emitting element is $H1$, and a distance from a side surface of one second light-blocking structure away from one corresponding second light-emitting element to the corresponding second light-emitting element is $H2$, wherein $|H1-H2|>0$.

16. The display panel according to claim 15, wherein:

$$S1>S2 \text{ and } H1>H2.$$

17. The display panel according to claim 1, wherein: along the first direction, a distance from one first light-blocking structure to one corresponding first light-emitting element is $d01$; along a second direction, a distance from the first light-blocking structure to the corresponding first light-emitting element is $d02$; and $d01=d02$, wherein the second direction and the first direction intersect in a direction parallel to the plane where the display panel is located.

18. The display panel according to claim 17, wherein: a shape of the orthographic projection of one first light-emitting element on the substrate is a circle, and a shape of the orthographic projection of one corresponding first light-blocking structure on the substrate is a ring, wherein the circle and the ring are a concentric structure.

19. The display panel according to claim 1, wherein: the first light-emitting element is a blue light-emitting element or a green light-emitting element, and the second light-emitting element is a red light-emitting element.

20. A display device, comprising a display panel, wherein the display panel includes: a substrate; light-emitting elements on a side of the substrate; and a light-blocking structure on a side of one light-emitting element away from the substrate, wherein: one light-emitting element includes a first light-emitting element and a second light-emitting element; an area of an orthographic projection of one first light-emitting element on the substrate is $S1$, and an area of an orthographic projection of one second light-emitting element on the substrate is $S2$; one light-blocking structure includes a first light-blocking structure and a second light-blocking structure; an orthographic projection of one first light-blocking structure on the substrate at least partially surrounds one corresponding first light-emitting element and a distance from the first light-blocking structure to the corresponding

first light-emitting element along a first direction is $d1$; an orthographic projection of one second light-blocking structure on the substrate at least partially surrounds one corresponding second light-emitting element and a distance from the second light-blocking structure to the corresponding second light-emitting element along the first direction is $d2$, wherein the first direction is parallel to a plane where the display panel is located; and $|S1-S2|>0$, and $|d1-d2|>0$.
